

Rachel Malea Russ

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EDUCATION

Purdue University, West Lafayette, Indiana
Bachelor's of Science in *Aeronautical and Astronautical Engineering*
Minors: Mathematics and Economics
Relevant Coursework:

- Fluid Dynamics, Aerodynamics, Introduction to Propulsion, Introduction to CFD, and Computational Combustion and Propulsion*

May 2027
GPA 3.66/4.00

Brentwood High School, Brentwood, Tennessee
Honors: Summa Cum Laude, GPA 4.63 weighted, 4.0/4.0 unweighted

May 2023

TECHNICAL SKILLS AND RESEARCH INTERESTS

Research Interests	Fluid Mechanics	Applied Mathematics	Numerical Methods / Computing
Research/ Technical Skills	Compressible Flow High-Speed / Hypersonic Flows Shock-Wave / Boundary-Layer Interaction Turbulent Flows Multiphase Flows PeleC Ansys Fluent	PDEs Numerical Analysis Computational Modeling Uncertainty Quantification	AMReX Adaptive Mesh Refinement HPC MPI Parallel Computing SLURM Python C Git/GitHub VisIt Ansys Mechanical

RESEARCH EXPERIENCE

Research Intern, **University of Tennessee Space Institute**, Tullahoma TN
with Dr. Esteban Cisneros-Garibay

- Investigate **hypersonic inlet unstart and shock-train dynamics** using computational fluid dynamics (CFD)
- Execute high-fidelity simulations using the **PeleC compressible flow solver** on high-performance computing clusters
- Adapt PeleC source code and simulation parameters to **investigate effects of operating conditions, numerical schemes, and mesh refinement strategies**
- Post-process large-scale simulation datasets using Python** and VisIt to analyze shock interactions, flowfield evolution, and inlet performance
- Assess **adaptive mesh refinement (AMR)** behavior and **numerical stability** to improve solution accuracy and diagnose numerical instabilities
- Communicated research findings through **technical presentations and scientific visualizations**

June 2026 - Current

Undergraduate Researcher, **Purdue University**, West Lafayette IN
with Dr. Haifeng Wang

- Develop **compressible CFD models** in Ansys Fluent to investigate **shock-wave propagation and steady shock behavior**
- Analyze pressure, velocity, and shock structure data to characterize compressible flow behavior

February 2026 - May 2026

OTHER EXPERIENCE

Propulsion/ Fluids Engineer, **Purdue Space Program (PSP)**, West Lafayette IN

- Supported fluid system design for an entry in the International Lander Challenge
- Designed waterline routing and produced piping and instrumentation diagrams (P&IDs) for fluid system documentation
- Contributed to **designing rocket tank, diffuser, and ground support equipment (GSE)**

August 2024 - May 2026

Intern, Vanderbilt LifeFlight, Nashville TN

June 2025 - August 2025

- Developed an HTML/JavaScript application that **displayed real-time helipad turbulence conditions for flight operations**
- Helped improve operational awareness for pilots and dispatch personnel
- **Integrated wind tunnel data** from RWDI with established operational turbulence thresholds
- Collaborated with engineering and aviation personnel to **improve flight safety decision-making**

Experimental Fluid Dynamics and Aerodynamics Laboratory,
Purdue University, West Lafayette IN

Spring 2025, Fall 2025

- **Operated subsonic and supersonic wind tunnels** to conduct aerodynamic and fluid dynamics experiments
- Processed experimental data to **evaluate flow behavior and aerodynamic performance**
- **Designed, proposed, and executed** two team-based experimental research projects
- Prepared technical reports for experimental findings

Structural Engineer, Purdue Design Build Fly, West Lafayette IN

August 2023 - May 2024

- Designed and manufactured structural components for an RC aircraft competing in the AIAA DBF competition
- **Collaborated with a multidisciplinary team** to design the aircraft fuselage in SolidWorks
- Fabricated wooden aircraft structures and validated manufacturability through iterative design

CAMPUS AND COMMUNITY SERVICE

Mentee, Women in Engineering at Purdue, West Lafayette IN

August 2023 - Current

- Participates in networking events and seminars that support the success of women in engineering
- Develops professional communication and networking skills through mentorship and career development programs

Volunteer, Adventure Science Center, Nashville TN

May 2024 - August 2024

- Taught and supervised children through hands-on STEM activities, including microscopy and woodworking
- Collaborated with fellow volunteers to create engaging educational experiences

Resident, Global Engineering LC, West Lafayette IN

August 2023 - December 2023

- Collaborated with students from diverse backgrounds to develop a global perspective and strengthen intercultural communication skills
- Engaged in discussions and activities that strengthened cross-cultural communication and examined engineering challenges from an international perspective

CERTIFICATIONS AND LANGUAGES

- **Certifications:** SolidWorks CAD Design Associate (2024), and CornellX Introduction to Engineering Simulations (FEA & CFD using Ansys) (2025)
- **Language:** German (Intermediate Proficiency)